

CEN202 Programming Languages Final Homework

Due 21.06.2020 until 23:59

Solve the below questions by yourself (it is NOT A GROUP HOMEWORK) and write your results by using Word or any word processing software. Upload .doc, .docx, or .pdf document to the system. Handwritten homework will not be accepted.

QUESTIONS

1. **(20 points)** Which characteristics of programming languages do you think are the most important and why? Write your own opinion.
2. **(30 points)** Design a state diagram to recognize variable names in a new language such that a variable name starts with any one of the % or @ or \$ or _ characters, then followed by at least one letter, then followed by any combination of letters and digits. Both lowercase and uppercase letters can be used. As examples; while %a and @x1Yz are valid variable names; \$1 and a2bc23xYz are not valid variable names. After that, write BNF or EBNF rules by using your state diagram to generate the valid variable names in the new language. Show that your rules can generate a valid variable name by applying a leftmost derivation.
3. **(30 points)** What is computed in the below code? What is the condition that the row satisfies? Explain briefly. Then, rewrite this code without goto in C language. After that, compare readability of your code to that of the code given below.

```
for (i=0; i < n; i++){
    for (j=0; j < n; j++){
        if (x[i][j]!=0)
            goto reject;
    }
    printf("First row that satisfies the condition is %d\n", i);
    break;
reject:
}
```

4. **(20 points)** Using the structures parent(X,Y), male(X), and female(X), write structures that define father(X,Y), mother(X,Y), brother(X,Y), sister(X,Y), grandparent(X,Y), and grandmother(X,Y).